

Session 2 — Lecture 0.2: Introduction to Functions

Core Set — 5 to 7 Questions

Q1. Find the domain of:

$$f(x) = \frac{5x - 2}{\sqrt{x^2 - 9}}$$

Q2. Find the domain and range of:

$$g(x) = \sqrt{16 - x^2}$$

Think geometrically before jumping to algebra.

Q3. Determine whether each function is even, odd, or neither. Show the algebraic test — don't just state the answer.

(a) $f(x) = x^4 - 5x^2 + 3$

(b) $g(x) = x^5 - 4x$

(c) $h(x) = x^3 + x^2 + 1$

Q4. Given the piecewise function:

$$f(x) = \begin{cases} x^2 + 1 & \text{if } x < 0 \\ 3x - 2 & \text{if } x \geq 0 \end{cases}$$

Find $f(-4)$, $f(0)$, $f(5)$. Then state the domain and describe the range.

Q5. A function is defined as:

$$f(x) = \frac{\sqrt{x+5}}{x^2 - x - 6}$$

Find the domain. Write your answer in interval notation.

Q6. Without a calculator, evaluate and simplify the difference quotient:

$$\frac{f(x+h) - f(x)}{h}, \quad f(x) = 3x^2 - 2x$$

Leave your answer fully simplified. Do not cancel h prematurely.

This one is important — the difference quotient is the definition of the derivative. You are already doing calculus here.

Q7. Embodied AI Reflection

A robotic arm operates using a function that maps joint angle θ as input to end-effector height y as output. In one sentence, explain why domain restriction is not just a mathematical formality but a physical necessity in this system.

Extension Set — When You Have Free Time

Q8. Find the domain of:

$$f(x) = \ln(x^2 - 4x + 3)$$

Recall that $\ln$ is only defined for strictly positive inputs. Solve the inequality carefully.

Q9. Given $f(x) = 2x^2 - 3$ and $g(x) = x + 4$, find:

(a) $f(g(2))$ (b) $g(f(x))$ (c) $f(f(x))$

This previews Lecture 0.4 on composition. Good to get an early feel for it.

Q10. A function satisfies $f(x) = f(-x)$ for all x . Its domain is $[-5, 5]$ and it has exactly one zero at $x = 0$.

- (a) What symmetry does this function have?
- (b) Sketch a rough description of what its graph must look like
- (c) Give one example of such a function and verify it satisfies all three conditions

Q11. Prove that the sum of two odd functions is always odd. Use the formal definition, not an example.

Challenge Question — Flow State Only

Q12. Consider:

$$f(x) = \frac{\sqrt{4 - x^2}}{\sqrt[3]{x^2 - 1}}$$

Find the complete domain. This requires you to handle a square root restriction AND a cube root that introduces its own complication simultaneously. Write your final answer as a union of intervals.

Think carefully about what the cube root allows versus what the square root restricts. They don't behave the same way.